

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electrical and Electronics Engineering
Academic Year: 2020 – 2021 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. EEE

Course Code & Title: EE8402 Transmission and Distribution

Name of the Faculty member: D. Karthik Prabhu

ACTIVITY

Topic: Types of Conductors in Transmission and Distribution System

Goal (Learning Outcome):

- ❖ The students will be able to explain various types of conductors used in transmission and distribution system - (To achieve this learning outcome the activity was chosen)

Use of appropriate methods:

Activity Chosen: One minute paper

Justification for Choosing the Activity for the topic:

1. Explain various types of conductors

The topic types of conductors have 3 types of in transmission and distribution conductors. Since each type has its own definition and limitation. After teaching the concept, I thought of conducting this activity for making the students to give the difference between the each type of conductors which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 4 Minutes

Effective presentation:

Details of the Implementation:

After teaching the concept, give students one or two minutes to think about the topic without writing anything.

Total Strength is 3

Reporter: Myself

At the end the Class (Last 5 minutes)

- ✓ I asked the students to think about various types of conductors in transmission and distribution for 2 minutes.
- ✓ Then I told them to write as much as they can within a short period of time (1 minute)
- ✓ Finally, I collected the papers from each column (1 minute)

Samples:

Type of standard conductor :-

K. Mubish Akash Varun.
953619105016.

AAC - All Aluminium

conductor.

AAAC - All Aluminium with alloy conductor.

ACSR - Aluminium conductor with steel reinforcement.

ALAR - Aluminium conductor with alloy reinforcement.

S. Abhinaya Sri

953619105001

Types of conductors :

ii-year : EEE

* Standard conductors : ~~So~~ This conductors has two or more conductors join together to pass high current.

- All aluminium conductors
- All aluminium and alloy conductors
- All Aluminium with steel reinforcement
- All Aluminium with alloy reinforcement

* Bundle conductors : This consists of more than two sub conductors connected parallelly which is used to reduce voltage drop in the conductor.

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt when told most of the points.
- ✓ While conducting the activity, I understood that the students will be able to explain various types of conductors in transmission and distribution.
- ✓ The success of the activity was evaluated by asking the same question in **Internal Assessment test I Retest – All the students answered.**

Reflective Critique:

1. Pre-implementation Reflection :

- **Benefits:**
 - Students can able to attend the question even in the questions are in indirect form.
 - Students can able to explain the concepts in examination without any confusion.
- **Challenges:**
 - Time utilization for conducting activity.

2. Post-implementation Reflection :

- **Benefits:**
 - Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.
- **Challenges:**
 - Slow learners were not able to understand some topics during discussion hours.

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C211.1	3	3	3	3	2	-	-	-	-	2	-	3

References:

1. R.K. Rajput, 'A Textbook of Power System Engineering', Laxmi Publications.
2. C.L.Wadhwa, 'Electrical Power System', New Age International Publishers.

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ACTIVITY

Topic: Skin Effect and Proximity Effect

Goal (Learning Outcome):

- ❖ The students will be able to explain the difference between skin effect and proximity effect in transmission and distribution system - (To achieve this learning outcome the activity was chosen)

Use of appropriate methods:

Activity Chosen: Think – Pair - Share

Justification for Choosing the Activity for the topic:

2. Differentiate skin effect and proximity effect

The topic skin effect and proximity effects are used to assess the performance of the transmission and distribution system. After teaching the concept, I thought of conducting this activity for making the students to give the knowledge about skin and proximity effect on performance of transmission line which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 6 Minutes

Effective presentation:

Details of the Implementation:

After teaching the concept of skin and proximity, the students were made to pair with their neighbours (i.e) In a Class; three students will be seated in a desk.

Total Strength is 33, Number of Pairs – 8

Photographer: Myself

Reporter: Myself

At the end the Class (Last 6 minutes)

- ✓ I asked the students to **think** about Skin effect and Proximity effect concept for 2 minute.

- ✓ Then I told them to **Pair** with their neighbours and discuss about the difference between Skin effect and Proximity effect for another 1 minute.
- ✓ Finally, I selected on Pair from each column randomly and ask them to **share** one difference between Skin effect and Proximity effect. (3 minutes)
- ✓ 3 Column: 3 Pairs gave one difference.
- ✓ Finally, I summarized the points and I told one missed point to all the students.

A view of implementation of the Activity while the students are sharing the points



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt when told most of the points.
- ✓ While conducting the activity, I understood that the students will be able to explain various types of conductors in transmission and distribution.

Reflective Critique:

3. Pre-implementation Reflection :

• Benefits:

- Students can able to attend the question even in the questions are in indirect form.
- Students can able to explain the concepts in examination without any confusion.

• Challenges:

- Time utilization for conducting activity.

4. Post-implementation Reflection :

• Benefits:

- Students understood the concept which was reflected from their answers during sharing session.

• Challenges:

- Slow learners were not able to understand some points during sharing session.

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C211.1	3	3	3	3	2	-	-	-	-	2	-	3

References:

1. V.K.Mehta, 'Principles of Power System', S. Chand Publications.
2. A. Chakrabarti, M.L.Soni, P.V.Gupta and U.S. Bhatnagar, 'A Text Book on Power System Engineering', DhanpatRai& Co.